The Thermodynamics of Generative AI

Progress Report: From 1D Dynamics to 2D Manifold Learning

Diffusion Model Development Team

May 26, 2026

Roadmap & Current State

14-Week Objective: Build a Diffusion Model from scratch.

- **[Completed]** Block 1: 1D Langevin Dynamics & VP-SDE.
- **[Completed]** Block 2: 2D Manifold Learning (Spiral Dataset).
- **[Up Next]** Block 3: High-Dimensional Generation (U-Net & MNIST).
- **[Future]** Block 4: The Analytical Limit & Bayesian Frameworks.

Today's Focus: Scaling the underlying thermodynamics from scalars to vector fields.

Block 2: The 2D Jump & The Manifold Hypothesis

Moving to a 2D geometry (a spiral) fundamentally breaks naive models due to the **Manifold Hypothesis**.

- **The Problem:** A 2D spiral is a 1D manifold embedded in 2D space.
- **Undefined Score:** If $\mathbf{x}$ is slightly off the manifold, $p_0(\mathbf{x}) = 0$.
- Therefore, the target $\nabla_{\mathbf{x}} \log p_0(\mathbf{x})$ is undefined for 99% of the space.

The Thermodynamic Solution

The forward SDE acts as a thermodynamic smoother. By injecting Gaussian noise, we expand the distribution's support, ensuring $p_t(\mathbf{x}) > 0$ everywhere, allowing the network to learn a continuous vector field.

Architectural Upgrade: Beating Spectral Bias

Challenge: Neural networks suffer from spectral bias, naturally learning low-frequency functions. In 1D, concatenating scalar time t was sufficient. In 2D, the score field changes too violently near $t \to 0$ for a scalar to provide enough signal.

Solution: *Sinusoidal Positional Embeddings.* We projected t into a high-dimensional spatial tensor:

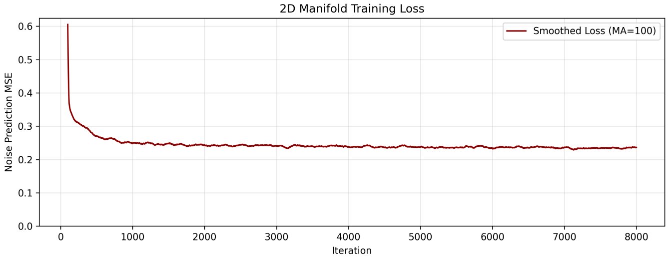
$$\mathbf{e}(t)_{2i} = \sin\left(\frac{t}{10000^{2i/d}}\right), \quad \mathbf{e}(t)_{2i+1} = \cos\left(\frac{t}{10000^{2i/d}}\right)$$

This is the exact time-conditioning mechanism used in modern Transformers and U-Nets, forcing the MLP to recognize high-frequency temporal shifts.

Vector-Valued Denoising Score Matching

Our loss function scaled perfectly from scalars to vectors. We minimized the variance-weighted squared L_2 norm over the 2D output:

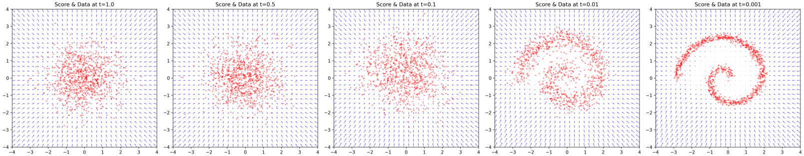
$$\mathcal{L}(\theta) = \mathbb{E}_{t,\mathbf{x}_0,\epsilon}\left[\sigma_t^2 \left\| \mathbf{s}_\theta(\mathbf{x}_t, t) + \frac{\epsilon}{\sigma_t} \right\|_2^2\right]$$



Results: Visualizing the Vector Field

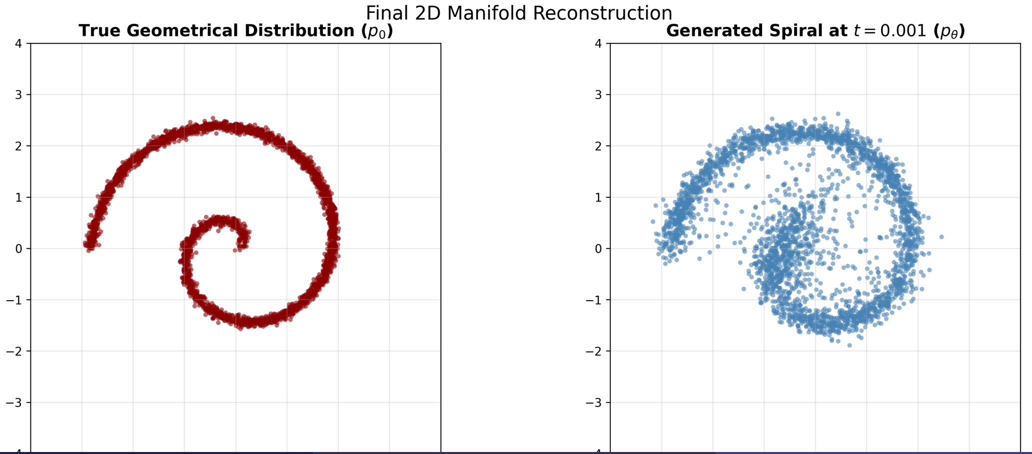
By generating a Quiver plot, we verified that the neural network learned the correct thermodynamic forces.

- At $t \to 1$, vectors point toward the origin ($\mathcal{N}(\mathbf{0}, \mathbf{I})$).
- As $t \to 0$, vectors aggressively pull noise onto the 1D spiral manifold.



Results: 2D Reverse Generation

Integrating the learned score field via our 2D Anderson SDE solver successfully reconstructed the complex spiral geometry from pure noise.



Next Steps

Block 2 is Complete.

We have successfully built, trained, and visualized a continuous-time Diffusion Model on complex geometric manifolds.

Next Up - Block 3:
Upgrading from MLPs to Convolutional U-Nets to handle high-dimensional image tensors (MNIST).